

build-cshared.sh

Cross-compile the lava-api-go embedded server surface (internal/mobile) into Android-loadable native libraries (liblavaapi.so) via `go build -buildmode=c-shared`, per ABI. This is the c-shared + JNI path chosen after gomobile bind was proven BLOCKED on this module (see [build-aar.sh.md](#) "KNOWN BLOCKER" — gomobile's overlay-module generator cannot resolve lava-api-go's relative replace `../submodules/*` directives). Unlike gomobile, `go build -buildmode=c-shared` uses ordinary Go module resolution, so it HONORS the replace directives.

Script: lava-api-go/scripts/build-cshared.sh Sources: lava-api-go/cmd/lavaapi-cshared/main.go (cgo c-shared entry), lava-api-go/cmd/lavaapi-cshared/jni/jni_bridge.c + .../jni/CMakeLists.txt (JNI bridge for Phase C/D). Output: lava-api-go/build/jniLibs/<abi>/liblavaapi.{so,h} (gitignored — never committed; only the SOURCE is committed).

What it does

1. Resolves the NDK toolchain bin dir, probing darwin-arm64 → darwin-x86_64 → linux-x86_64 (on Apple Silicon macs the prebuilt dir is still named darwin-x86_64, run via Rosetta).

2. For each requested ABI, sets CC to the matching NDK clang wrapper and runs:

```
CGO_ENABLED=1 GOOS=android GOARCH=<arch> CC=<ndk-clang> \  
GOMAXPROCS=2 nice -n 19 \  
go build -buildmode=c-shared -o build/jniLibs/<abi>/  
liblavaapi.so ./cmd/lavaapi-cshared
```

3. Prints each .so's size, sha256, and file(1) type.

4. Verifies the exported symbols with the NDK `llvm-nm -D` and asserts LavaApiStart is present in the dynamic symbol table.

ABI → toolchain mapping

ABI	GOARCH	NDK clang wrapper (API=28 default)	file reports
arm64-v8a	arm64	aarch64-linux-android28-clang	ELF 64-bit ARM aarch64
x86_64	amd64	x86_64-linux-android28-clang	ELF 64-bit x86-64
	arm		

ABI	GOARCH	NDK clang wrapper (API=28 default)	file reports
armeabi-v7a		armv7a-linux-androideabi28-clang	ELF 32-bit ARM EABI5

Usage

```
cd lava-api-go
./scripts/build-cshared.sh                # all three ABIs
./scripts/build-cshared.sh arm64-v8a     # just the critical ABI
```

Environment overrides (all optional)

Variable	Default	Purpose
ANDROID_NDK_HOME	~/Library/Android/sdk/ndk/25.1.8937393	NDK root for the cgo cross-compile
ANDROID_API	28	API level baked into the clang wrapper name

Exit codes

Code	Meaning
0	all requested ABIs built; size + sha256 + symbols verified
1	one or more ABIs failed (bad ABI, missing clang, build error, missing symbol)
2	configuration error (no NDK toolchain bin dir, no llvm-nm)

Exported C surface

The cgo entry (cmd/lavaapi-cshared/main.go) exports four plain C functions (declared in the generated liblavaapi.h). All string args/returns are NUL-terminated UTF-8; every returned char* is heap-allocated by C.CString and MUST be released by the caller via LavaApiFree:

```
extern char* LavaApiStart(char* configJSON); // "" on success,
      else error msg
extern char* LavaApiStop(void);              // "" on success,
      else error msg
extern char* LavaApiStatus(void);            // status JSON
      document
extern void LavaApiFree(char* p);            // free a returned
      string
```

configJSON is {"bindAddr":"0.0.0.0","port":8443,"sqlitePath":"/data/x.db"} (bindAddr/port optional, default 0.0.0.0:8443; sqlitePath required). The underlying lifecycle is internal/mobile.{Start,Stop,Status}, serving the FULL production Gin router over TLS, SQLite-backed.

JNI bridge — the contract Phase C MUST match

cmd/lavaapi-cshared/jni/jni_bridge.c adapts the flat C ABI to the JNI naming convention. The Kotlin side (Phase C) binds exactly:

```
package digital.vasic.lava.apigo
```

```
object LavaNative {
    external fun nativeStart(configJson: String):
        String // "" on success, else error message
    external fun nativeStop():
        String // "" on success, else error
        message
    external fun nativeStatus(): String //
        status JSON document
}
```

The JNI symbol names the bridge defines (package dots → underscores, then class, then method):

- Java_digital_vasic_lava_apigo_LavaNative_nativeStart
- Java_digital_vasic_lava_apigo_LavaNative_nativeStop
- Java_digital_vasic_lava_apigo_LavaNative_nativeStatus

A Kotlin object compiles to a final class with a static INSTANCE; its external functions register as native methods that receive a jclass, so the bridge functions take (JNIEnv*, jclass, ...).

jni_bridge.c + its CMakeLists.txt live in the jni/ SUBDIRECTORY (not the cgo main package) on purpose: jni_bridge.c #includes liblavaapi.h, the cgo-generated header this build PRODUCES, so it cannot be compiled by the cgo build itself. It is compiled later by the Android NDK toolchain via externalNativeBuild { cmake { path = "../jni/CMakeLists.txt" } }. Set LAVAAPI_PREBUILT_DIR (defaults to lava-api-go/build/jniLibs); ANDROID_ABI selects the per-ABI subdir holding liblavaapi.{so,h}.

Verification (real evidence)

For each ABI the script confirms, and you can re-confirm manually:

```
file lava-api-go/build/jniLibs/arm64-v8a/liblavaapi.so
~/Library/Android/sdk/ndk/25.1.8937393/toolchains/llvm/prebuilt/
darwin-x86_64/bin/llvm-nm \
-D lava-api-go/build/jniLibs/arm64-v8a/liblavaapi.so | grep
LavaApi
```

You CANNOT run an android/arm64 .so on a macOS host — symbol-presence + file-type verification is the honest proof at this layer. Actual JNI invocation is proven later by the Phase E on-device Challenge.

Constitutional notes

- §6.T.2 — the cgo cross-compile is resource-limited (GOMAXPROCS=2 nice -n 19).
- §6.U — no sudo/su; the NDK clang + llvm-nm are user-local tools.
- §6.R — NDK path + API level come from env vars with discovery fallbacks; no host-specific path is hardcoded.
- §6.J / **Anti-Bluff** — the script reports REAL build output. A failed build or a missing exported symbol is reported as a failure with verbatim output; no .so is faked.
- §11.4.18 — this file is the mandated external doc for the script.